

Auteur: Siebe Haché
Studierichting: Informatica
Oefeningenreeks 3A nr. 36.17

$$\begin{aligned} & \frac{d}{dx} (\ln (x^2 + \sqrt{x^4 - 1})) \\ &= \frac{1}{x^2 + \sqrt{x^4 - 1}} \cdot \left(2x + \frac{4x^3}{2\sqrt{x^4 - 1}} \right) \\ &= \frac{1}{x^2 + \sqrt{x^4 - 1}} \cdot \frac{2x\sqrt{x^4 - 1} + 2x^3}{\sqrt{x^4 - 1}} \\ &= \frac{2x(\sqrt{x^4 - 1} + x^2)}{(x^2 + \sqrt{x^4 - 1}) \cdot \sqrt{x^4 - 1}} \\ &= \frac{2x}{\sqrt{x^4 - 1}} \end{aligned}$$

References